On Market Making for a New Exchange: EUR/USD Designated Market Maker on Exchange A

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1 Introduction

1.1 Motivation

A newly launched exchange — referred to throughout this report as exchange A — has mandated our firm as the sole Designated Market Maker for its EUR/USD spot book. Two established venues, B and C, remain the primary sites of price discovery, carrying on average 75% and 25% of the daily volume respectively. We receive market data from B with a 200 ms delay and from C with a 170 ms delay, while high-frequency participants reach the same data in 50 ms and can act on exchange A before our own view of fair value has refreshed. The mandate commits us to continuously quoting ten levels on each side of the book at a common 1 bp tick, with a starting capital K split evenly between EUR and USD and an asymmetric fee grid. A delta limit caps the EUR inventory we are willing to warehouse before externalising risk as taker flow on B and C.

Three constraints shape every design decision that follows. First, on the day exchange A goes live no historical fills or counterparty behaviour are observable, so the initial quoting policy cannot be learnt from data and must be derived from first-principles microstructure models. Second, the latency gap to HFTs exposes every resting quote to adverse selection, which the pricing model must compensate for rather than ignore. Third, because we cannot market-make on B or C, the delta limit turns inventory management into an active cross-venue hedging problem rather than a passive book-balancing exercise.

1.2 Objective

Our goal is to deliver an end-to-end market-making system that addresses the three constraints above and can be evaluated under the conditions described in the subject brief. The project is split into three phases matching the mandate timeline. Phase 1 bootstraps liquidity without any trading history, using a heuristic quoting policy derived from classical market-making theory. The fair mid is reconstructed from the two reference venues at their respective latencies, the bid-ask spread is sized to compensate for both inventory risk and adverse selection, and quotes are skewed in the direction of recent order-flow imbalance. A dedicated hedge router externalises excess EUR inventory on B and C whenever the delta limit is approached. Phase 2 re-estimates the Phase 1 parameters by maximum likelihood from a simulated month of fills. Phase 3 studies how our quoting policy should degrade from a price-competitive quoter to a reliable fallback liquidity provider once aggressive HFT market makers populate the inside of the book.

1.3 Simulation Environment

Because the exchange has no history at launch and we cannot run the strategy against real counterparties, the evaluation environment is a layered simulation built specifically for this project. A stochastic mid-price generator (GBM, GARCH with jumps, or Heston) produces the fair value underlying both reference venues. A venue simulator wraps that path into a realistic bid/ask and depth profile per venue, with a choice of five spread regimes (static, stochastic, adaptive, asymmetric, and skewed) reflecting the microstructure assumptions we want to stress. A limit order book with a matching engine and fill events sits at the centre of exchange A; a Poisson client-flow generator with truncated Pareto sizes delivers counterparty orders whose intensity decays exponentially in

the quoted half-spread; and a reporting layer reconstructs the full mark-to-market and realised P&L decomposition from the trade history.

1.4 Evaluation

Each phase is evaluated on the subject-mandated one-month simulated backtest, with the capital K reset to its initial split at the start of every phase. For Phase 1 we report realised and mark-to-market P&L, the percentile envelope of mark-to-market as a function of time since trade inception, top-ten trades by size, full-versus-partial fill rates at each of the ten quoting levels, and inventory trajectories against the delta limit. For Phase 2 we compare the heuristic and calibrated configurations on an independent month using the same diagnostics, so that any difference in performance is attributable to the updated parameters rather than to the market regime. For Phase 3 we measure both our own P&L under HFT competition and the quality of liquidity observed by exchange-A clients, distinguishing small from large trades and quantifying the continuity of two-sided quoting through volatility spikes, one-sided HFT retreats, and venue outages.

1.5 Report Structure

The remainder of the report follows the step-by-step structure of the subject. section 2 introduces the order-book data structure and matching rule; section 3 describes the venue simulator and the family of spread regimes supported; section 4 presents the quoting framework in full; section 5 covers submission and cancellation handling; section 6 documents the simulation loop that binds the modules together; section 7 details the inventory and P&L bookkeeping; section 8 gathers the Phase 1 backtest outputs; section 9 analyses the calibration results; section 10 examines the Phase 3 fallback behaviour; and section 11 summarises the findings and outlines the remaining modelling questions.

2 Order Book

The order book of exchange A must show a price and a quantity at ten levels on both sides at all times. This section describes the data structure we chose to hold that state, how submissions and cancellations flow through it, and the invariants of the matching engine. The design is exercised on a realistic EUR/USD seed (mid = 1.10, 1 bp tick, ten levels per side, 1M EUR per level).

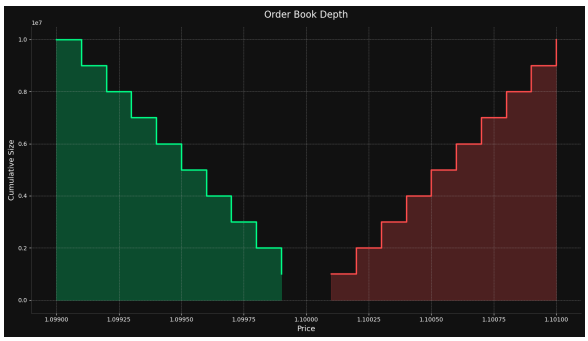
2.1 Data structure

Two pressures shape the design. Matching requires a price-time-ordered view of the market-maker (MM) side at clearing time, yet the hot path is dominated by bulk requotes — up to ten cancels followed by ten posts per step. Keeping the book permanently sorted would pay on every mutation for a benefit consumed only once per step.

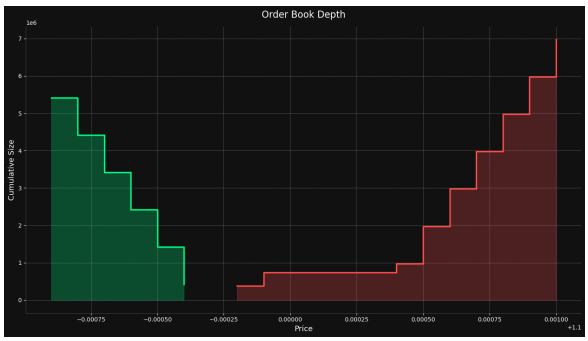
We therefore split orders by origin (MM versus client) into two hash-indexed stores and keep a price-ordered view of the MM side as a lazy cache, rebuilt only when the matcher runs. Cancels and posts touch only the hash stores; the MM side is sorted once per clearing call instead of being re-ordered on every mutation. Time priority within a price level is preserved by a counter that grows by one on every insertion, so later orders carry higher numbers than earlier ones.

2.2 Submissions and cancellations

The two sides are handled asymmetrically. MM cancels and posts arrive in bulk and never run the matcher, because passive quotes cannot cross by construction. Client orders are inserted one at a time, and the matcher runs immediately after each insertion.



(a) Seeded book: one order per level, symmetric depth.



(b) After twenty marketable client orders: shallow levels consumed from the top.

Figure 1: Cumulative depth before and after a burst of client flow.

2.3 Matching rule

Three invariants govern every clearing call. Client orders only match against MM orders, never against each other: as the sole designated market maker for Phases 1 and 2 we are on one side of every trade. The fill price is always the passive MM price, never the incoming aggressive price — clients pay the spread we quote, which is the only source of gross revenue. And matching follows strict price-time priority with partial fills: the engine walks the sorted MM side, deducts the minimum of the two sizes, logs the match,

and either retires the filled order or leaves the remnant resting. The residual shape in Figure 1 is a direct visual check of this rule: the top of the book is consumed first, deeper levels are untouched.

2.4 Fills as events

Every match is dispatched synchronously, as a fill event, to a single registered listener. In the full simulation the listener is the Quoter, which updates inventory, trade history, and the pending top-up queue inside the clearing call. The Quoter therefore never polls the book: the next quote computation already sees the post-fill inventory, which removes staleness bugs on steps that both fill and requote. Decoupling the book from its consumer also keeps the same engine usable in Phase 3, where a separate diagnostic listener records fills for the exchange-client-side liquidity analysis without touching the Phase 1 code path.

3 Market Simulator for Venues B and C

Exchanges B and C are external reference venues: we never post on them, we only read them. Our quoting engine on A consumes the best bid, the best ask, and the best-size of each, and nothing else. We therefore do not simulate a full limit order book on B or C: all the depth beyond the top of book would be detail the rest of the system never looks at. Instead we simulate a mid-price process and build a synthetic bid/ask/depth layer around it. This is the second of the two options offered by the subject, and it is both cheaper to run and sufficient for everything the market maker on A has to decide. The reduced-form choice keeps realism where it matters — spreads that widen under stress and depth that thins with them — while avoiding the combinatorial cost of simulating individual order arrivals at two venues.

3.1 Noised mid

The fundamental value V_t is the GBM (or GARCH, or Heston) path produced by the stock simulator. The observed mid on B and C is V_t plus microstructure noise, in the spirit of Roll [4]:

$$P^{\text{mid}}_{B/C}(t) = V_t + \eta \cdot \sigma \cdot \sqrt{\Delta t} \cdot Z_t, \quad Z_t \sim \mathcal{N}(0,1),$$

with $\eta = 0.1$. Each venue draws an independent Z_t path, so B and C are noised versions of the same fundamental. One subtle point: every spread model below estimates realized volatility from V_t, not from P^{mid}, because differencing the noised series would double-count the microstructure variance inside the vol estimator. The zero-mean estimator used for that (see `compute_rv_zero_mean`) is unbiased for zero-drift GBM and lower-variance than the sample std over short windows.

3.2 Spread regimes

We benchmarked five regimes of increasing realism — Static, Stochastic, Adaptive, Asymmetric, and Skewed — and kept two. We reject Static because a price-independent width ignores volatility entirely; the market maker is widest exactly when they should be widest, but never adapts. Stochastic scales the width with realized vol but never returns to the competitive floor even in calm markets, which systematically over-charges. Asymmetric mean-reverts realistically but treats up-moves and down-moves the same way, leaving a latency-arbitrage gap for HFTs who, seeing the vol spike 150 ms before us, can infer our quote path. We deploy **Adaptive on** C and **Skewed on** B.

Venue C — **Adaptive.** A symmetric half-spread proportional to how much realized vol deviates from the long-run input σ:

$$h(t) \;=\; h_0 \cdot \Big(1 + \alpha \cdot \big(\widehat{RV}_t/\sigma - 1\big)\Big), \qquad h(t) \geq 0.1 \cdot h_0,$$

with $h_0 = \text{mid}(t) \cdot s_{\text{bps}}/10000$ and $\widehat{RV}_t$ an EMA-smoothed rolling vol over a 120 s window. The width is tight at baseline, widens on shocks, and tightens symmetrically when calm returns. A 10% floor prevents collapse near zero. C carries 25% of volume — a smaller venue where informed flow is weaker, competition enforces a tight floor, and the symmetry of up vs. down adjustment is not a material leak.

Venue B — Skewed. Same vol-adaptive width, plus a directional shift of the quote centre driven by a short-horizon momentum signal. Let $\hat{m}_t$ be the EMA of log-returns normalised by its steady-state noise floor; the skew is

$$\delta_t \;=\; \gamma \cdot h(t) \cdot \tanh\bigl(\mathrm{sign}(\hat{m}_t) \cdot \max(|\hat{m}_t| - \tau,\, 0)\bigr),$$

with $\tau = 1.5$ the dead-band threshold. Small random wiggles map to $\delta_t = 0$ exactly; only sustained directional moves shift the centre. Posted quotes become ask $=$ mid $+ h + \delta$ and bid $=$ mid $- h + \delta$: width is unchanged, the centre moves in the direction of the trend. B carries the dominant 75% of volume, where short-horizon order-flow imbalance predicts returns [2]; a momentum-unaware venue there would be repeatedly picked off [1, 3].

3.3 Depth

Depth at each step is derived deterministically from spread width, not simulated as an independent process:

$$\mathrm{depth}_t \;=\; \bar{D} \cdot \frac{\overline{S}}{S_t},$$

where S_t is the full spread and $\overline{S}$ its session mean. The economic story is simple: when the external market maker is confident, they quote tight and large; when stressed, they pull size and widen. The hedge router on A reads this array with the same 200 ms (B) / 170 ms (C) latency offset as the mid, and caps each hedge leg at the available depth.

3.4 One stock, two venues

B and C are two `Market` objects wrapping the same `Stock` via `copy.deepcopy`. The fundamental path V_t is shared; the noise draws and spread realisations are independent. We deliberately do not model B/C price divergence: the cross-venue edge in this project is the latency asymmetry (HFTs see B/C at 50 ms while we see them at 200/170 ms), not price disagreement between venues.

3.5 Validation

Figure 3.5 shows the five width paths under a GARCH-plus-jump run — the one regime where the models visibly differentiate. Each width panel is annotated with its Pearson correlation to realized vol; the bottom panel shows the skew δ_t activating at each jump under the Skewed regime and remaining at zero elsewhere.

[Figure placeholder]
spread-regimes-garch.png — export from
`market_garch.compare_spreads()` in `test/price_simulation.ipynb`

Figure 2: Width panels with realized-vol overlay and the Skew δ_t path, GARCH+jump run.

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References

[1] Bruno Biais, Pierre Hillion, and Chester Spatt. An empirical analysis of the limit order book and the order flow in the Paris bourse. *The Journal of Finance*, 50(5): 1655–1689, 1995.

[2] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

[3] Robert F. Engle and Andrew J. Patton. Impacts of trades in an error-correction model of quote prices. *Journal of Financial Markets*, 7(1):1–25, 2004.

[4] Richard Roll. A simple implicit measure of the effective bid-ask spread in an efficient market. *The Journal of Finance*, 39(4):1127–1139, 1984.